ASEN 5090 HW1

Assigned: 26 August 2026

Due: 4 September 2026

1. (M&E Book Homework Problem 1-4)

A *pseudolite* (short for pseudo-satellite) consists of a generator of a GPS-like signal and a transmitter. Pseudolites are used to augment the GPS signals. Suppose an observer is constrained to be on the line joining two pseudolites PL1 and PL2, which are separated by 1 km. The pseudolite clocks are perfectly synchronized but the observer's clock may have an unknown bias with respect to the pseudolite clocks. Estimate the observer's position and clock bias given that the pseudoranges to PL1 and PL2 are (a) 550 m and 500 m, respectively, and (b) 400 m and 1400 m, respectively.

2. Measurement Geometry

A ground station located at x_G=0, observes an aircraft flying by at constant speed (v0) at constant altitude (h0), traveling straight from horizontal position –x0 to +x0, passing directly above the ground station location. The ground station is able to make measurements of the zenith angle (z), range (R), and range rate ($\dot{R}$) to the aircraft. We are interested in studying how useful these measurements are for determining the x-position of the aircraft (x_A).

a) Derive expressions for each of the measurements - range, range rate, and zenith, as a function of x_A , and the parameters v0 and h0.
b) Write a simple, well-organized program/code to simulate the aircraft motion for specified values of v0, x0, and h0; and for this motion, compute and plot the range, range rate, and zenith angle measurements based on the expressions you found in part a. Plot the measurements for the following values: v0=50m/s, x0=250m, h0=100 m.

Now, assume that the aircraft height (h0) is known perfectly, and the measurements are to be used to estimate the horizontal position of the aircraft, x_A.

c) How useful are the three types of measurements for determining x_A? In other words, how sensitive is each measurement to the aircraft position? For which portion(s) of the flight does each measurement work well and where is it not very useful? Explain.
d) Does the sensitivity/utility of the observations get better if the aircraft is flying at a lower altitude? Explain.
e) (Optional – How would an error in the assumed height affect your results?)

3. M&E Book Problem 2-1

4. M&E Book Problem 2-2